

Networking and Content Delivery

Where traffic enters, what decides whether it is allowed, and how it reaches distant users.

Three questions: where does traffic **enter**, what **decides** whether it is allowed, and how does it **reach** distant users?

Amazon VPC

- ◆ **A logically isolated section of the AWS Cloud** where you control the network layout.
- ◆ **A VPC belongs to one Region. Each subnet belongs to one Availability Zone.**
- ◆ Route tables decide where traffic from a subnet is sent.
- ◆ The VPC itself costs nothing. Some components inside it, such as NAT gateways, do.
- ◆ Nothing in a VPC is reachable from the internet unless you deliberately arrange it.

ON THE EXAM

A subnet sits in one Availability Zone. Spanning zones always means more than one subnet.

Public or private subnet?

PUBLIC SUBNET

Has a route to the internet gateway

- Route table sends 0.0.0.0/0 to an internet gateway
- Resources can have public IP addresses
- For load balancers, NAT gateways, bastion hosts
- Reachable from the internet if security allows

PRIVATE SUBNET

Has no such route

- No direct route to an internet gateway
- For application servers and databases
- Outbound internet only via a NAT gateway
- Never directly reachable from the internet

THE TIEBREAKER

A subnet is public or private purely because of its route table. Nothing else about it differs.

Gateways

- ◆ An **internet gateway** allows two-way traffic between a VPC and the internet.
- ◆ A **NAT gateway** lets private resources reach the internet outbound, while blocking inbound connections.
- ◆ The NAT gateway sits in a public subnet; the private subnet routes through it.
- ◆ NAT gateways are charged per hour and per gigabyte processed, so they appear in cost questions.
- ◆ A NAT gateway is not a security control. It permits outbound traffic; it does not filter it.

ON THE EXAM

Instances must download patches but must not be reachable from the internet is the NAT gateway scenario, every time.

Security group or network ACL?

SECURITY GROUP

Instance level, stateful

- Attaches to an instance's network interface
- Return traffic is allowed automatically
- Allow rules only; there is no deny
- All rules evaluated together

NETWORK ACL

Subnet level, stateless

- Attaches to a subnet
- Return traffic needs its own explicit rule
- Allow and deny rules both available
- Rules evaluated in numbered order

THE TIEBREAKER

Instance means security group. Subnet means network ACL. Needing to block a specific address means network ACL, because security groups cannot deny.

VPC endpoints

- ◆ An endpoint lets resources in a VPC reach an AWS service without traversing the internet.
- ◆ **Gateway endpoints** serve Amazon S3 and DynamoDB, and carry no additional charge.
- ◆ **Interface endpoints**, powered by AWS PrivateLink, serve most other services and are charged.
- ◆ Traffic stays on the AWS network, which satisfies requirements about data not crossing the internet.
- ◆ They can also remove the need for a NAT gateway when the only outbound traffic is to AWS services.

ON THE EXAM

Must not traverse the public internet is the endpoint signal. It appears in both security and cost questions.

Amazon Route 53

- ◆ **Three jobs: domain registration, DNS routing, and health checking.**
- ◆ Health checks let it route traffic away from endpoints that have stopped responding.
- ◆ **It is a global service**, not a Regional one.
- ◆ It routes to AWS resources and to endpoints outside AWS equally well.
- ◆ It resolves names to addresses. It does not carry your application traffic.

ON THE EXAM

Domain registration, DNS routing and health checks together is the Route 53 phrase used almost verbatim on the exam.

Route 53 routing policies

POLICY	USE IT WHEN
Simple	One record, one destination. No decision to make
Weighted	Split traffic by proportion. Useful for testing a new version
Latency	Send users to the Region that responds fastest for them
Failover	Send to a primary, and to a standby if health checks fail
Geolocation	Route based on where the user is, often for compliance or language

ON THE EXAM

Fastest for the user means latency. A percentage split means weighted. Where the user is means geolocation.

Amazon CloudFront

- ◆ **Caches content at edge locations** so users fetch it from nearby instead of from the origin.
- ◆ The origin can be Amazon S3, an Application Load Balancer, or a server outside AWS.
- ◆ **It reduces latency and also reduces load on the origin**, because cached requests never reach it.
- ◆ AWS WAF and AWS Shield Standard both operate at the CloudFront edge.
- ◆ Signed URLs and signed cookies restrict access to private content.

ON THE EXAM

Distant users, slow page loads and static content together are the CloudFront signature.

CloudFront or Global Accelerator?

AMAZON CLOUDFRONT

Caches content

- Stores copies at edge locations
- HTTP and HTTPS content
- Best for static and cacheable content
- Users get a cached copy, not the origin

AWS GLOBAL ACCELERATOR

Improves the path

- No caching at all
- Any TCP or UDP traffic, not just HTTP
- Routes over the AWS backbone from the nearest edge
- Provides static IP addresses that never change

THE TIEBREAKER

Cacheable content means CloudFront. Non-HTTP protocols, static IPs, or dynamic traffic that cannot be cached means Global Accelerator.

Direct Connect or Site-to-Site VPN?

AWS DIRECT CONNECT

A dedicated private link

- Physical connection, bypassing the internet
- Consistent, predictable bandwidth and latency
- Takes weeks to provision
- Higher cost, for sustained heavy use

SITE-TO-SITE VPN

Encrypted over the internet

- Set up in hours, over your existing connection
- Encrypted, but performance varies with the internet
- Low cost, quick to establish
- Often used as a backup to Direct Connect

THE TIEBREAKER

Consistent and predictable means Direct Connect. Quick, cheap and encrypted means VPN. Time to provision often decides.

Transit Gateway and API Gateway

- ◆ **AWS Transit Gateway is a network hub** connecting many VPCs and on-premises networks through one point.
- ◆ It replaces a tangle of individual VPC peering connections as the number of VPCs grows.
- ◆ **Amazon API Gateway creates, publishes and manages APIs** for applications to call.
- ◆ It handles throttling, authorisation and versioning, and commonly fronts AWS Lambda.
- ◆ They share a word and nothing else. One connects networks; the other publishes APIs.

ON THE EXAM

Many VPCs needing to talk to each other means Transit Gateway. Publishing an API to callers means API Gateway.

Reverse lookup

IF THE SCENARIO SAYS

THE ANSWER IS

Instances need patches but must not be reachable inbound

NAT gateway in a public subnet

Block one specific IP address

Network ACL

Restrict access to a specific instance

Security group

Traffic to S3 must not cross the public internet

VPC gateway endpoint

Users worldwide, slow static content

Amazon CloudFront

Non-HTTP traffic, static IP addresses needed

AWS Global Accelerator

Consistent bandwidth to our data centre

AWS Direct Connect

Connect forty VPCs to each other

AWS Transit Gateway

ON THE EXAM

Eight phrasings covering most networking questions. Protocol and predictability words do most of the work.

A company runs application servers that must download operating system patches from the internet. Security policy states that nothing on the internet may open a connection to those servers.

The servers currently sit in a private subnet with no internet access at all.

What should the company add?

- ☐ A An internet gateway attached to the private subnet
- ☐ B A NAT gateway in a public subnet, with the private subnet routing to it
- ☐ C A security group rule allowing inbound traffic on port 443
- ☐ D Public IP addresses on the application servers

◆ PAUSE AND DECIDE

What should the company add?

- A

 An internet gateway attached to the private subnet
That would make it a public subnet, allowing inbound connections. It breaks the stated policy.
- B

 A NAT gateway in a public subnet, with the private subnet routing to it
Outbound patching works; inbound connections are impossible. Exactly the requirement.
- C

 A security group rule allowing inbound traffic on port 443
Opens inbound access, which is precisely what the policy forbids.
- D

 Public IP addresses on the application servers
Makes the servers directly addressable from the internet. The opposite of the requirement.

THE SIGNAL

Outbound yes, inbound no, is the NAT gateway definition. Three of the four options all open inbound access.

A security team has identified a single IP address that is repeatedly probing a company's public web servers. They want to block all traffic from that address before it reaches any instance in the web subnet.

The web servers already have security groups permitting HTTP and HTTPS from anywhere.

How should the team block that address?

- ☐ A Add a deny rule to the instances' security group
- ☐ B Add a deny rule to the subnet's network ACL
- ☐ C Remove the inbound HTTP rule from the security group
- ☐ D Attach an internet gateway rule blocking the address

◆ PAUSE AND DECIDE

How should the team block that address?

- A

 Add a deny rule to the instances' security group
Security groups have no deny rules. This option describes something that cannot be configured.
- B

 Add a deny rule to the subnet's network ACL
Network ACLs support deny rules and act at the subnet boundary, before traffic reaches any instance.
- C

 Remove the inbound HTTP rule from the security group
Blocks the whole internet, not one address. It takes the site offline.
- D

 Attach an internet gateway rule blocking the address
Internet gateways have no filtering rules. Another invented capability.

THE SIGNAL

Blocking one address is the one thing a security group cannot do. That alone selects the network ACL.

A media company serves large video files and images from Amazon S3 to viewers across Europe, Asia and South America. Viewers far from the bucket's Region report slow starts, and the S3 data transfer bill is rising.

The content is identical for every viewer and changes infrequently.

What should the company use?

- ☐ A Amazon CloudFront
- ☐ B AWS Global Accelerator
- ☐ C S3 buckets replicated into each Region
- ☐ D Route 53 latency-based routing

◆ PAUSE AND DECIDE

What should the company use?

- A** Amazon CloudFront
Caches identical content at edge locations near viewers. Latency falls and origin transfer falls with it.
- B** AWS Global Accelerator
Improves the path but caches nothing, so every request still reaches S3 and the bill does not fall.
- C** S3 buckets replicated into each Region
Would help latency, but multiplies storage cost and adds replication to manage.
- D** Route 53 latency-based routing
Routes users to the fastest endpoint, but there is only one bucket, so there is nothing to choose between.

THE SIGNAL

Identical, infrequently changing content is cacheable. Cacheable plus a rising transfer bill means CloudFront.

A gaming company runs servers in three Regions. Players connect over UDP and the company must publish two fixed IP addresses that never change, because players' clients have them hard-coded.

It wants players routed to the healthiest, nearest Region automatically.

Which service meets these requirements?

- | | |
|--|---|
| ☐ A Amazon CloudFront | ☐ B AWS Global Accelerator |
| ☐ C An Application Load Balancer in each Region | ☐ D Route 53 geolocation routing |

◆ PAUSE AND DECIDE

Which service meets these requirements?

- A

 Amazon CloudFront
HTTP and HTTPS only, and caching is meaningless for live game traffic. UDP rules it out immediately.
- B

 AWS Global Accelerator
Static anycast IP addresses, UDP support, and automatic routing to the nearest healthy Region.
- C

 An Application Load Balancer in each Region
ALBs handle HTTP, do not provide fixed IPs, and would not choose between Regions.
- D

 Route 53 geolocation routing
Routes by where the user is, not by health or latency, and DNS gives names rather than fixed IPs.

THE SIGNAL

UDP plus fixed IP addresses is the Global Accelerator pair. Either one alone would already eliminate CloudFront.

A bank runs applications in a VPC that write large volumes of data to Amazon S3. Compliance requires that this traffic never traverse the public internet. The applications currently reach S3 through a NAT gateway, and the NAT charges are significant. The bank wants to satisfy compliance and reduce cost.

What should the bank implement?

- ☐ A An AWS Direct Connect link
- ☐ B A VPC gateway endpoint for Amazon S3
- ☐ C A second NAT gateway for redundancy
- ☐ D A Site-to-Site VPN to Amazon S3

◆ PAUSE AND DECIDE

What should the bank implement?

- A

 An AWS Direct Connect link
Connects an on-premises network to AWS. Both ends here are already inside AWS.
- B

 A VPC gateway endpoint for Amazon S3
Traffic stays on the AWS network, satisfying compliance, and gateway endpoints carry no charge.
- C

 A second NAT gateway for redundancy
Adds cost and still routes traffic out through the internet path.
- D

 A Site-to-Site VPN to Amazon S3
VPNs connect networks to a VPC. You do not build a VPN to an AWS service.

THE SIGNAL

Both requirements point one way. Endpoints keep traffic on the AWS network and gateway endpoints are free.

A manufacturer needs to connect its factory network to AWS. Data must move continuously with predictable latency for a control application, and the company has budget for dedicated infrastructure.

It also wants a lower-cost connection in place within a week, to begin testing before the main link is ready.

Which two should the manufacturer use? (Choose two.)

- ☐ A AWS Direct Connect
- ☐ B AWS Site-to-Site VPN
- ☐ C AWS Transit Gateway
- ☐ D Amazon CloudFront
- ☐ E A VPC peering connection

◆ PAUSE AND DECIDE

Which two should the manufacturer use? (Choose two.)

- A** AWS Direct Connect
Predictable latency and dedicated bandwidth for the control application, once provisioned.
- B** AWS Site-to-Site VPN
Established in hours over the existing internet connection, so testing can start within a week.
- C** AWS Transit Gateway
A hub for connecting many VPCs. Only one network is being connected here.
- D** Amazon CloudFront
Content delivery to end users. Nothing to do with connecting a private network.
- E** A VPC peering connection
Connects two VPCs to each other. It cannot connect an on-premises network.

THE SIGNAL

Two timescales in one question: within a week points at VPN, predictable latency points at Direct Connect.

A company has grown to forty VPCs across several teams, and they increasingly need to communicate with one another and with the corporate data centre. The network team is maintaining a large and growing number of individual peering connections. It wants a single place to manage this connectivity.

Which service should the company adopt?

- ☐ A More VPC peering connections
- ☐ B AWS Transit Gateway
- ☐ C Amazon API Gateway
- ☐ D AWS PrivateLink for each VPC pair

◆ PAUSE AND DECIDE

Which service should the company adopt?

- A

 More VPC peering connections
Peering does not scale. Every new VPC needs a connection to every other, which is the problem described.
- B

 AWS Transit Gateway
A single hub. Each VPC and the data centre connect once, and routing is managed centrally.
- C

 Amazon API Gateway
Publishes APIs. It shares a word with Transit Gateway and nothing else.
- D

 AWS PrivateLink for each VPC pair
Exposes a specific service privately. It is not a general network connectivity solution between VPCs.

THE SIGNAL

Growing number of individual connections is the phrase. Transit Gateway replaces a mesh with a hub.

A company is releasing a new version of its website and wants to send ten percent of visitors to it, keeping ninety percent on the current version, so it can compare error rates before committing.

Both versions run behind their own load balancers in the same Region.

Which Route 53 routing policy should the company use?

- | | |
|--|--|
| ☐ A Latency-based routing | ☐ B Failover routing |
| ☐ C Weighted routing | ☐ D Geolocation routing |

◆ PAUSE AND DECIDE

Which Route 53 routing policy should the company use?

A Latency-based routing

Chooses the fastest endpoint for each user. Both are in the same Region, so there is no latency difference to exploit.

B Failover routing

Sends traffic to a standby only when the primary is unhealthy. Both versions here are healthy and wanted.

C Weighted routing

Splits traffic by proportion, which is exactly a ten and ninety percent split.

D Geolocation routing

Routes by the user's location. Location is irrelevant to a version comparison.

THE SIGNAL

A stated percentage split is always weighted routing. The same-Region detail exists to eliminate latency routing.

What to carry out of this module

- ◆ **A subnet is public or private purely because of its route table.** Nothing else differs.
- ◆ **Outbound yes, inbound no, means a NAT gateway.** Blocking one address means a network ACL.
- ◆ Must not cross the public internet means a VPC endpoint. Gateway endpoints for S3 and DynamoDB are free.
- ◆ **CloudFront caches HTTP content. Global Accelerator improves the path for any protocol.**
- ◆ Predictable means Direct Connect. Quick and cheap means VPN. They are often used together.

ON THE EXAM

Transit Gateway connects networks. API Gateway publishes APIs. They share a word and nothing else.